PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	COVID-19 and Health Care Workers: A rapid systematic review into risks and preventive measures
AUTHORS	Gross, J.; Mohren, Judith; Erren, Thomas

VERSION 1 – REVIEW

REVIEWER	Atanu Sarkar Memorial University, Canada
REVIEW RETURNED	24-Jul-2020

GENERAL COMMENTS	Occupational hazards for health care workers due to COVID-19 is very important issue. The authors did a rapid systematic review of risks encountered by the HCWs from COVID-19, prevention and suggestion for future research agenda. Since, the disease is novel, it is understandable to get good quality peer reviewed article in such a short time. However, the manuscript is well written. However, there are some issues the authors are expected to address. 1. In the introduction section, it is desirable to add some global estimated figures on number of HCWs got infection and died. Probably the best way to staart with the Chinese whistle blower who died eventually, to draw more interest. 2. in Page 14, the authors (of the manuscript) should confirm from the articles, how the authors of the selected articles made sure that the HCWs contacted infection from workplace not from own family members or other non-occupational sources. If there is no such confirmation, the authors of the manuscript should mention this as weakness of the studies. 3. In page 14 the authors mentioned of 'insufficient PPE'. Does it mean insufficient number and also lack of quality assurance (such N95 vs general surgical mask etc.) 4. Is there any indication of categories of HCWs more at risk than others? Rahman et al (Middle Eastern Respiratory Syndrome Coronavirus Infections in Saudi Arabia) found HCWs having significantly less fatality, probably due to access to health care, better knowledge and early diagnosis (doi: 10.2105/AJPH.2019.305186). 5. For mental health the authors should ensure that the articles used unified mental health evaluation tools. But, it is unlikely. Therefore this can be another weakness. Probably Germany and China have different mental health assessment techniques. If yes, it should be highlighted. 6. Since most of the articles are of low quality, how to deal with the results and importance in policy perspectives. 7. Cost benefit ratio is not clear. Needs rewriting.

REVIEWER	Walter Zingg Infection Control Programme and WHO Collaborating Center on Patient Safety University of Geneva Hospitals, Geneva, Switzerland
REVIEW RETURNED	18-Aug-2020

GENERAL COMMENTS	The authors present the results of a systematic review on various aspects on the exposure of healthcare workers to COVID-19 patients with a number of associated outcomes. On the positive side, the authors timely identified the potential challenges around professional exposure of healthcare workers to COVID-19 during the pandemic; on the other side, they rushed too early into too a broad search exercise, which in the end produced a heterogeneous picture of limited concepts rather than actionable results. Comments The topics addressed by the authors are relevant for anticipating future strategies in preventing cross-transmission in healthcare settings and in implementing best practice procedures with personal protective equipment (PPE). My main concern is that the review focused on published studies until 26 April 2020 only. Over the past months, we learned quite a bit on how to address the challenges on exposing healthcare workers to COVID-19 patients. The main problem, as it seems now, was unpreparedness, and exposure outside healthcare settings played an important role. Still, substantial controversy remains on correct use of PPE and how healthcare workers get infected (if at all). Contrary to the view of the authors, the limited time period of the review remains a limitation and not a strength in my opinion. In addition, most information available at that time came from Mainland China, and generalisation to other countries may not be appropriate. The heterogeneous findings of the review appear more like a mapping exercise than a systematic review focusing on a specific mechanism and outcome. Together with the short time period of the review, the results go to too many directions. I think, based on this work, the authors should focus now on one of several areas: what are the risk factors of healthcare workers to get SARS-CoV-2? What is the outcome of healthcare workers with COVID-19 compared to the general population? What are the somatic side effects on wearing PPE? What are the psychological consequences of being exposed to COVID-19 patients professionally? These are all valid questions where the manuscript as is only offers general information and hypotheses rather than actionable results. On the method side, while the authors defined the study questions, they were less explicit about what methodology would be acceptable to address the study question: e.g. for the study question about risks for infection in association of COVID-19 exposure, the best methodology would have been case-control studies, or controlled cohort studies; for psychological outcomes, the best method would be qualitative research applying interviews or validated questionnaires. Now, the review includes qualitative research as well as quantitative with or without control, questionnaires, case series, and uncontrolled cohort studies.

	Some abbreviations, e.g. PTSD and many psychological tests in the table, are not explained. Abstract: the abstract is not very informative and does not include all PRISMA elements. Study design: although the study design per se is valid and the authors seemed to proceed accordingly, the broadness of the anticipated search resulted in limited results and conclusions. Searching for one topic for a longer time period would have been more useful. Limitations: only the short period of time is mentioned by the authors and presented as a strength rather than limitation. As mentioned above, I would have expected more rigorous inclusion criteria. Supplementary reporting: the authors should mention and cite the PRISMA guidelines in the text. As for the latter, the authors were quite generous in mapping out the PRISMA elements to their manuscript.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1	**Authors' comment**	**Changes done**
1. In the introduction section, it is desirable to add some global estimated figures on number of HCWs got infection and died. Probably the best way to start with the Chinese whistle blower who died eventually, to draw more interest.	We do understand that the reviewer is interested in numbers of infected HCW. Hence, we have included some information. However, including information about individual persons, like the whistleblower, that are controversially discussed, do not match with the intention of this paper.	Health care workers [HCW] – as important actors of the critical infrastructure – are under considerable risk of infection and other diseases. Within the first months of the pandemic (until early March), 3300 HCWs in China were described as infected, in Italy about 20% of responding HCW[2]. Because of the risks for HCW, in our review, we ask (…)
2. in Page 14, the authors (of the manuscript) should confirm from the articles, how the authors of the selected articles made sure that the HCWs contacted infection from workplace not from own family members or other non-occupational sources. If there is no such confirmation, the authors of the manuscript should mention this as weakness of the studies.	Thank you. We included this important information in the discussion section.	Moreover, as the virus is not specific for workplaces, and therefore HCWs could get infected outside the workplace, the studies should assess the risk of infection in private life, too. Unfortunately, these non-workplace associated risks were not explicitly considered.
3. In page 14 the authors mentioned of 'insufficient PPE'. Does it mean insufficient number	We understand that the wording is not exhaustive. We checked the original publication that was	improper PPE

and also lack of quality assurance (such N95 vs general surgical mask etc.)	cited here and changed the wording according to the wording the authors used (instead of “insufficient”, now: ”improper”). The authors did not specify the word “improper”.	
4. Is there any indication of categories of HCWs more at risk than others? Rahman et al (Middle Eastern Respiratory Syndrome Coronavirus Infections in Saudi Arabia) found HCWs having significantly less fatality, probably due to access to health care, better knowledge and early diagnosis (doi: 10.2105/AJPH.2019.305186).	This is a highly interesting question. However, we focused on the risk of getting infected as well as on preventive measures, but not on the outcome of the infection. Therefore, we cannot answer this question.	None
5. For mental health the authors should ensure that the articles used unified mental health evaluation tools. But, it is unlikely. Therefore this can be another weakness. Probably Germany and China have different mental health assessment techniques. If yes, it should be highlighted.	Yes, we agree. We included this information in table 1 and 2. However, as it was included in the column referring to “outcome” it was not that obvious. Therefore, we changed the column’s title from: “Outcome” to: “outcome / evaluation tool”. In the discussion, we had already mentioned that the different outcome assessments have a negative impact on comparing the studies (page 16, lines 21-23): “For this review, we found great heterogeneity in self-administered outcome assessment (Tables 1 and 2),	Outcome/evaluation tool
6. Since most of the articles are of low quality, how to deal with the results and importance in policy perspectives.	Ultimately, it is interesting and relevant to draw conclusions for policy. However, our review is from a scientific perspective. We discuss that recent knowledge is limited and that better studies are needed. Moreover, we present suggestions to improve studies concerning COVID-19. We are convinced that during the pandemic for political decisions, medical insights like from such reviews, are crucial. However, other perspectives – like economical, sociological or legal – have certainly to be considered, too. While our review may be(come) part of an	None

	"insight base" to meet the reviewer's sensible expectation, it would not be sufficient to draw conclusions for policy from this systematic literature "snapshot".	
7. Cost benefit ratio is not clear. Needs rewriting.	OK. We think that we missed to clarify what we meant with "costs" and "benefits". We did not focus on costs such as financial aspects but more on the comparison between what the workers invested (like work or extra time or inconvenience or risk regarding other health facets (such as headaches)) and the benefit from the prevention measure, such as reducing risks of getting infected. Stating what we mean with costs and benefits will allow to understand.	When discussing prevention measures, possible negative effects have to be considered and investigated (…), the effectiveness of the specific measure (the benefit) has to be weighed up against possible risks and inconveniences (the costs).
Reviewer 2		
My main concern is that the review focused on published studies until 26 April 2020 only. Over the past months, we learned quite a bit on how to address the challenges on exposing healthcare workers to COVID-19 patients. The main problem, as it seems now, was unpreparedness, and exposure outside healthcare settings played an important role. Still, substantial controversy remains on correct use of PPE and how healthcare workers get infected (if at all). Contrary to the view of the authors, the limited time period of the review remains a limitation and not a strength in my opinion.	In the "strengths and limitation section" of the review we explictitly stated the time period as a limitation: **"Limitations:** • The literature focuses on the first phase (until 26. April 2020) of the pandemic." However, the "rapid review" format is certainly not only a limitation – see also below.	None
In addition, most information available at that time came from Mainland China, and generalisation to other countries may not be appropriate.	We definitely agree. Therefore, we present the countries where studies were conducted in table 1 and 2. Moreover, we have now	However, as demonstrated in this review, high-quality studies from different parts of the world are lacking. Therefore, we do agree with Glasziou that the current

	included "geographical bias" in the discussion.	phase of the pandemic has generated replicative studies of low methodological quality highlighting the need for well-defined study questions and robust study designs in order to reduce waste in COVID-19 research[56]. Moreover, studies from different countries and economical settings are needed to avoid "geographical bias". To achieve this end (…)
The heterogeneous findings of the review appear more like a mapping exercise than a systematic review focusing on a specific mechanism and outcome. Together with the short time period of the review, the results go to too many directions. I think, based on this work, the authors should focus now on one of several areas: what are the risk factors of healthcare workers to get SARS-CoV-2? What is the outcome of healthcare workers with COVID-19 compared to the general population? What are the somatic side effects on wearing PPE? What are the psychological consequences of being exposed to COVID-19 patients professionally? These are all valid questions where the manuscript as is only offers general information and hypotheses rather than actionable results.	Yes, this is a rapid systematic review. Clearly, it is a snapshot of first weeks/months into a highly unique challenge – no less but also no more. Thus, it is a documentation of initial reactions by scientists to understand, and possibly prevent, risks for HCWs. The fact that we are witnessing further "waves" of infection and disease worldwide makes our recommendations continue to be timely. We aimed to give an overview about the recent studies concerning our precise study questions. Due to the limited quality of the studies as well as the rapidly changing recommendations, presenting actionable results concerning risk of infection or specific prevention measures would not appear appropriately justified. But as a specific result from our search, we present suggestions how further studies could be planned and conducted to improve research towards (more) interpretable and sustainable results. The additional questions the reviewer is presenting are highly interesting, but would require other papers.	None
On the method side, while the authors defined the study questions, they were less explicit about what methodology would be acceptable to address the study question: e.g. for the study question about risks for infection	Yes, this is due to the fact that we wanted to give an overview – see above: a systematic snapshot of HCW-associated research or of "immediate-response-research" regarding a worldwide crisis.	None.

in association of COVID-19 exposure, the best methodology would have been case-control studies, or controlled cohort studies; for psychological outcomes, the best method would be qualitative research applying interviews or validated questionnaires. Now, the review includes qualitative research as well as quantitative with or without control, questionnaires, case series, and uncontrolled cohort studies.		
Some abbreviations, e.g. PTSD and many psychological tests in the table, are not explained.	We explained the abbreviation PTSD within the text and below the table now. All other abbreviations on different psychological tests that were presented in the table were already explained below the table. We checked this again and did not find any abbreviation missing.	PTSD: Post-traumatic stress disorder
Abstract: the abstract is not very informative and does not include all PRISMA elements.	The PRISMA elements are explained in the method section of the paper.	Now added: Overall, most studies on health risks as well as on the effectiveness of preventive measures were of a moderate to low quality; this was mainly due to limitations in study design, imprecise exposure and outcome assessments.
Study design: although the study design per se is valid and the authors seemed to proceed accordingly, the broadness of the anticipated search resulted in limited results and conclusions. Searching for one topic for a longer time period would have been more useful.	With interest we have read these suggestions. As outlined above, our “rapid systematic review” provides an informative “study snapshot” for an initial time segment of an unprecedented challenge worldwide. The relevant alternative suggestions by the reviewer could be viewed as developed “in hindsight”, i.e. after our rapid review’s synthesis of strengths and limitations of early HCW research and could now lead to differently targeted other papers.	None
Limitations: only the short period of time is mentioned by the authors and presented as a	See above	None

strength rather than limitation. As mentioned above, I would have expected more rigorous inclusion criteria.		
Supplementary reporting: the authors should mention and cite the PRISMA guidelines in the text. As for the latter, the authors were quite generous in mapping out the PRISMA elements to their manuscript.	OK. In the text, now we have included the information that we used the PRISMA-Statement.	We screened identified relevant studies according to the PRISMA-Statement (see figure 1).

VERSION 2 – REVIEW

REVIEWER	Atanu Sarkar Memorial University of Newfoundland, Canada
REVIEW RETURNED	27-Sep-2020

GENERAL COMMENTS	The authors addressed the outstanding issues. Now it is acceptable.